SPRiF: Spectral Projective Reset Integrate-and-Fire Neuron

Anonymous submission

Abstract

Temporal tasks with long delays require SNNs to retain information while continuing to communicate through spikes. Standard spiking neurons use the same membrane state to integrate context, generate spikes, and receive reset after a spike. Each spike can therefore disturb the temporal trace carried by that state. We introduce SPRiF (Spectral Projective Reset Integrate-and-Fire Neuron), a spectral construction within each neuron. SPRiF combines a constrained slow transition with one real decay mode and a damped rotational pair, a fast discharge state with two coordinates, and a learnable projective reset. The direct reset at the current timestep acts only on the fast state, while the slow state continues its spectral evolution. Across five temporal classification benchmarks, SPRiF has the highest reported mean accuracy among the compared methods on three tasks and remains competitive on the other two. Mechanism ablations identify the merged variant as the weakest configuration. On Spike-Intervention Delayed Match-to-Sample (SI-DMS), SPRiF maintains accuracy close to the clean condition at intervention counts beyond the training range. Analysis of impulse responses reveals heterogeneous learned temporal kernels. Together, these results support the tested SPRiF construction, which combines structured slow dynamics with direct reset of the fast discharge state.

Introduction

Spiking neural networks (SNNs) encode and transmit temporal information through discrete spike events. The underlying neuron dynamics must preserve relevant context while continuing to generate those events, a requirement that becomes difficult when dependencies span multiple timescales.

The leaky integrate-and-fire (LIF) neuron remains widely used because it is simple and compatible with training through surrogate gradients. Its extensions introduce adaptive variables or thresholds (Baronig et al. 2025; Zhang et al. 2025b), resonant dynamics (Higuchi et al. 2024; Huang et al. 2024; Zhang et al. 2025a), heterogeneous temporal decays (Wei et al. 2026; Deng et al. 2025; Fulleda-Garcia, Soldado-Magraner, and Margarit-Taulé 2026; Tiwari and Chauhan 2025), and modified reset (Guo et al. 2022; Huang et al. 2026; Li et al. 2025; Fang et al. 2023). These mechanisms broaden the available temporal responses, but the variable that generates spikes usually remains both the memory carrier and the reset target.

The reset operation therefore alters the same recurrence that transports input history. This coupling motivates assigning temporal memory and resettable discharge to distinct state groups while retaining binary communication and the same layer width.

We introduce **SPRiF** (Spectral Projective Reset Integrate-and-Fire Neuron), which implements this separation through two state groups. A slow spectral state combines one real exponential mode with a damped rotational mode represented by two coordinates. It stores temporal context. A fast discharge state projects this context into a membrane readout, emits binary spikes, and receives a learnable projective reset along $[1, \lambda]^\top$. SPRiF is trained with backpropagation through time and surrogate gradients. Reset acts directly on the fast state while the slow state continues its spectral evolution.

SPRiF assigns structured temporal evolution and correction after a spike to separate state groups. Its slow state stores temporal context through spectral modes, while its fast state handles spike readout and reset.

We evaluate SPRiF on five benchmarks spanning sequential images, physiological signals, speech represented as frames, and spoken digits represented as events. SPRiF has the highest reported mean accuracy among the compared methods on S-MNIST, PS-MNIST, and QTDB and remains competitive on GSC and SHD. Its parameter counts for the complete models are comparable to or lower than those of the strongest reported baselines. Mechanism ablations, recurrent spike intervention, and analyses of impulse responses examine the roles of rotation, state separation, and projective reset.

We make three contributions.

1. We introduce SPRiF, a spiking unit that maintains two state groups within each neuron. It combines a constrained slow transition with one real decay mode and one rotational pair, a fast discharge state with two coordinates, and a learned projective reset.
2. We establish bounded-input stability for the intrinsic slow transition and derive an exact identity for the reset at the current timestep, which acts only on the fast state.
3. We evaluate SPRiF on five temporal benchmarks and

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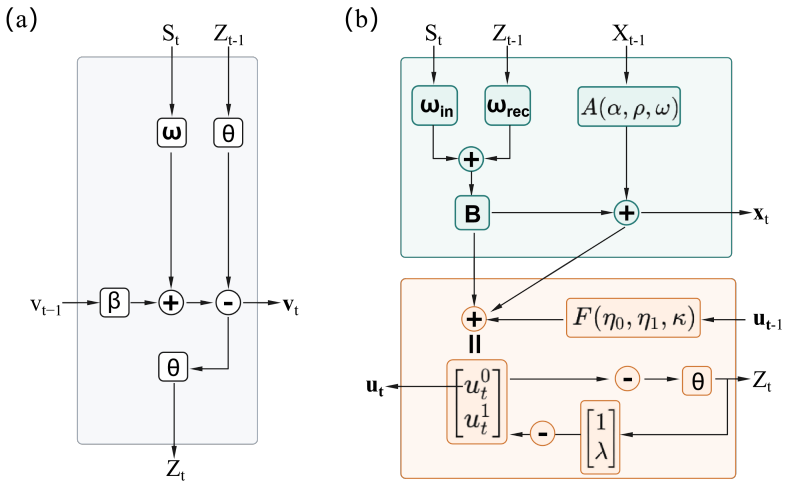


Figure 1: Architectures of (a) LIF and (b) SPRiF. SPRiF separates a slow memory state with no direct local reset update from a resettable fast discharge state.

examine its components through controlled ablations, SI-DMS, and analyses of temporal kernels.

Related Work

Neuron Dynamics for Temporal Processing

Adaptive variables, heterogeneous decays, and temporal gates broaden the response repertoire of LIF neurons (Yin, Corradi, and Bohté 2021; Baronig et al. 2025; Deng et al. 2025; Tiwari and Chauhan 2025). Resonate-and-fire variants encode damped oscillations (Izhikevich 2001; Higuchi et al. 2024; Huang et al. 2024; Zhang et al. 2025a), while heterogeneous neurons and synapses distribute temporal decays across a network (Perez-Nieves et al. 2021; Deng et al. 2025).

Compartmental and dual pathway SNNs assign computations related to memory and spike generation to distinct states or pathways (Zhang et al. 2024; Zheng et al. 2024; Pagkalos, Chavlis, and Poirazi 2023; Zhang et al. 2023; Sun et al. 2026; Zhang 2025). Within each neuron, SPRiF combines a constrained slow transition with one real decay mode and one rotational pair, a fast discharge state with two coordinates, and a learned projective reset. The direct reset at the current timestep acts only on the fast state.

Structured state-space dynamics provide another route to processing long temporal dependencies (Fabre et al. 2025; Shen et al. 2025). DMP-SNN uses memory shared across a layer and represented with few dimensions, together with a fast feedforward spiking pathway (Sun et al. 2026); SPRiF places both state groups within each neuron.

Reset Dynamics

Existing mechanisms soften, adapt, or remove the update after a spike. Recent work also revisits reset and refractory periods through recurrent sequence models with binary activations (Guo et al. 2022; Li et al. 2025; Huang et al. 2026; Fang et al. 2023; Zhang 2025). SPRiF applies learned projective reset to its fast discharge state. Strategies for temporal processing and robustness at the network level (Ding et al. 2025b, 2024, 2025a) complement this construction at the neuron level.

Methodology

Reset Coupling in Conventional LIF Dynamics

For presynaptic activities $s_{i,t}$, a standard leaky integrate-and-fire (LIF) neuron receives input current

$$I_t = \sum_i w_i s_{i,t} + b, \tag{1}$$

where w_i is the synaptic weight of input i and b is a bias term. Its membrane update is

$$v_t = \beta v_{t-1} + I_t - z_{t-1}\theta, \tag{2}$$
$$z_t = H(v_t - \theta). \tag{3}$$

Here, $\beta \in (0, 1)$ is the leak factor, H is the Heaviside step function, and θ is the firing threshold. The scalar v_t carries temporal context, determines spike timing, and receives the previous-step reset term. Unrolling over k steps gives

$$v_t = \beta^k v_{t-k} + \sum_{\ell=0}^{k-1} \beta^\ell \big(I_{t-\ell} - \theta z_{t-\ell-1}\big). \tag{4}$$

Input and reset histories are filtered through the same recurrence, so reset is inseparable from the state that transports temporal information. This motivates distinct memory and post-spike discharge states.

SPRiF Spiking Neuron Model

Figure 1 contrasts the two architectures. For layer input $\mathbf{s}_t$, the input current is

$$I_t = W_{\text{in}}\mathbf{s}_t + W_{\text{rec}}\mathbf{z}_{t-1} + b, \tag{5}$$

where the recurrent term is omitted in feedforward layers. In the equations for one neuron below, I_t denotes the scalar current received by that neuron. Each neuron j maintains a

slow state $\mathbf{x}_t \in \mathbb{R}^3$ with three coordinates and a fast state $\mathbf{u}_t \in \mathbb{R}^2$ with two coordinates; $\widetilde{\mathbf{u}}_t$ denotes the fast state before reset. Its dynamics are

$$\begin{cases} \mathbf{x}_t = A\mathbf{x}_{t-1} + BI_t, \\ \widetilde{u}_t^0 = \eta_0 u_{t-1}^0 + \kappa u_{t-1}^1 + (G\mathbf{x}_t)^0, \\ \widetilde{u}_t^1 = \eta_1 u_{t-1}^1 + (G\mathbf{x}_t)^1, \\ v_t = \widetilde{u}_t^0, \qquad z_t = H(v_t - \theta), \\ \mathbf{u}_t = \widetilde{\mathbf{u}}_t - z_t\theta \begin{bmatrix} 1 \\ \lambda_j \end{bmatrix}. \end{cases} \tag{6}$$

The update follows $I_t \to \mathbf{x}_t \to \widetilde{\mathbf{u}}_t \to v_t \to z_t \to \mathbf{u}_t$. The slow state is updated before spike generation; the fast state is then read out, thresholded, and reset. The local reset leaves the slow state unchanged at the current timestep, while $W_{\mathrm{rec}}\mathbf{z}_t$ can carry spike effects into later updates of the slow state in recurrent layers.

Equation (6) is written for one neuron, with indices suppressed except in λ_j. Except for W_{in} and W_{rec}, the listed dynamics parameters are per-neuron. Layers apply the update at each timestep, and recurrence is optional.

Spectral Dynamics of the Slow State

The update of the slow state in Eq. (6) uses a real-valued system with three coordinates. It contains one real decay mode and one complex-conjugate pair that produces rotation. This constrained transition retains temporal context while keeping its spectrum interpretable.

$$A = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \rho\cos\omega & -\rho\sin\omega \\ 0 & \rho\sin\omega & \rho\cos\omega \end{bmatrix}, \quad B = \begin{bmatrix} 1-\alpha \\ 1-\rho \\ 0 \end{bmatrix}. \tag{7}$$

The spectrum of A is

$$\mathrm{eig}(A) = \left\{\alpha, \rho e^{\mathrm{i}\omega}, \rho e^{-\mathrm{i}\omega}\right\}. \tag{8}$$

The first coordinate is an exponential trace, $x_t^0 = \alpha x_{t-1}^0 + (1-\alpha)I_t$, while (x_t^1, x_t^2) form a damped rotation driven through the first oscillatory coordinate. We parameterize

$$\alpha = \sigma(\alpha_{\mathrm{raw}}), \quad \rho = \sigma(\rho_{\mathrm{raw}}), \quad \omega = \pi\sigma(\omega_{\mathrm{raw}}), \tag{9}$$

where σ is the logistic sigmoid. Thus $\alpha, \rho \in (0,1)$ and $\omega \in (0,\pi)$ place every eigenvalue of A strictly inside the unit disk. The parameters α, ρ, and ω control real decay, rotational damping, and rotational frequency, respectively, yielding a stable, interpretable spectrum.

The resulting temporal basis is explicit. For a unit impulse entering the slow state, the response after k subsequent transitions is

$$A^k B = \begin{bmatrix} (1-\alpha)\alpha^k \\ (1-\rho)\rho^k\cos(k\omega) \\ (1-\rho)\rho^k\sin(k\omega) \end{bmatrix}. \tag{10}$$

The real coordinate supplies a monotone exponential trace, whereas the rotational pair supplies damped oscillations with different phases. They form a compact temporal basis with distinct decay and frequency controls.

Fast Discharge Dynamics and Projective Reset

The equations for the fast state in Eq. (6) read out slow context and implement resettable discharge. The learned coefficients $\eta_0, \eta_1 \in (0,1)$ are fast leaks, κ couples the second fast coordinate to the first, and $G \in \mathbb{R}^{2\times 3}$ projects slow dynamics into discharge space.

The first fast coordinate is the membrane potential $v_t = \widetilde{u}_t^0$, and the forward pass generates $z_t = H(v_t - \theta)$ with a hard threshold. During optimization, only the threshold derivative is replaced by the multi-Gaussian surrogate (Yin, Corradi, and Bohté 2021); the forward dynamics remain unchanged.

After a spike, SPRiF applies the following projective reset only to the fast state:

$$\mathbf{u}_t = \widetilde{\mathbf{u}}_t - z_t\theta \begin{bmatrix} 1 \\ \lambda_j \end{bmatrix}, \tag{11}$$

where λ_j is learned independently for neuron j. Setting $\lambda_j = 0$ recovers scalar subtraction on the membrane coordinate; otherwise, reset modifies both fast coordinates along a learned direction. Thus, λ_j controls the correction after a spike without changing the transition of the slow state.

Analysis of Reset and Memory Separation

The preceding equations separate the effect of reset at the current timestep from the later propagation of spike signals. Relative to the states before reset, the increments induced by reset are

$$\Delta\mathbf{x}_t^{\mathrm{reset}} = \mathbf{0}, \qquad \Delta\mathbf{u}_t^{\mathrm{reset}} = -z_t\theta \begin{bmatrix} 1 \\ \lambda_j \end{bmatrix}. \tag{12}$$

In LIF, the reset contribution $-z_{t-1}\theta$ is accumulated through the same recurrence as input history. In SPRiF, Eq. (12) gives a zero increment for $\mathbf{x}_t$ at the current timestep and confines reset to $\mathbf{u}_t$.

The recurrence of the slow state makes this distinction explicit. Iterating the first line of Eq. (6) yields

$$\mathbf{x}_t = A^t\mathbf{x}_0 + \sum_{\tau=1}^{t} A^{t-\tau}BI_\tau. \tag{13}$$

When the input current sequence is held fixed, Eq. (13) gives $\partial\mathbf{x}_t/\partial z_t = \mathbf{0}$ for the local reset operation. In recurrent layers, $\mathbf{z}_t$ can still affect $\mathbf{x}_{t+1}$ through $W_{\mathrm{rec}}\mathbf{z}_t$. The construction therefore distinguishes direct reset at the current timestep from later communication through the network.

Experiments

Evaluation Questions and Protocol

The experiments address four questions: how SPRiF compares with published temporal SNNs, which design choices contribute to its accuracy, whether forced updates triggered by spikes disrupt retained context, and what temporal dynamics the trained states learn. We evaluate five benchmarks spanning sequential images (S-MNIST and PS-MNIST), physiological signals (QTDB), speech represented as frames (GSC), and spoken digits represented as events (SHD). For SPRiF, Table 1 reports the mean and standard deviation over five seeds.

Dataset	Method	Network	Parameters (M)	Accuracy (%)
S-MNIST	$\text{GLIF}_{\text{NeurIPS, 2022}}$ (Yao et al. 2022)	Rec	0.15	96.64
	$\text{ASRNN}_{\text{Nat. Mach. Intell., 2021}}$ (Yin, Corradi, and Bohté 2021)	Rec	0.15	98.70
	$\text{DH-SNN}_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024)	Rec	0.08	98.90
	$\text{BRF}_{\text{ICML, 2024}}$ (Higuchi et al. 2024)	Rec	0.069	99.10
	$\text{TC-LIF}_{\text{AAAI, 2024}}$ (Zhang et al. 2024)	Rec	0.15	99.20
	DMP-SNN (II)$_{\text{Nat. Mach. Intell., 2026}}$ (Sun et al. 2026)	DMP	0.073	99.20
	SPRiF (ours)	**Rec**	**0.067**	**99.28 ± 0.06**
PS-MNIST	$\text{GLIF}_{\text{NeurIPS, 2022}}$ (Yao et al. 2022)	Rec	0.15	90.47
	$\text{ASRNN}_{\text{Nat. Mach. Intell., 2021}}$ (Yin, Corradi, and Bohté 2021)	Rec	0.15	94.30
	$\text{DH-SNN}_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024)	Rec	0.08	94.52
	$\text{BRF}_{\text{ICML, 2024}}$ (Higuchi et al. 2024)	Rec	0.069	95.20
	$\text{TC-LIF}_{\text{AAAI, 2024}}$ (Zhang et al. 2024)	Rec	0.15	95.36
	DMP-SNN (I)$_{\text{Nat. Mach. Intell., 2026}}$ (Sun et al. 2026)	DMP	0.061	95.50
	SPRiF (ours)	**Rec**	**0.067**	**95.86 ± 0.14**
QTDB	$\text{ASRNN}_{\text{Nat. Mach. Intell., 2021}}$ (Yin, Corradi, and Bohté 2021)	Rec	0.0018	85.90
	$\text{DH-SNN}_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024)	Rec	0.00178	86.35
	$\text{BRF}_{\text{ICML, 2024}}$ (Higuchi et al. 2024)	Rec	0.00173	87.00
	$\text{SE-adLIF}_{\text{Nat. Commun., 2025}}$ (Baronig et al. 2025)	Rec	0.0018	86.88
	SPRiF (ours)	**Rec**	**0.0017**	**88.43 ± 0.36**
GSC	$\text{SNN-SFA}_{\text{eLife, 2021}}$ (Salaj et al. 2021)	Rec	4.19	91.21
	$\text{ASRNN}_{\text{Nat. Mach. Intell., 2021}}$ (Yin, Corradi, and Bohté 2021)	Rec	0.31	92.10
	$\text{RadLIF}_{\text{Front. Neurosci., 2022}}$ (Bittar and Garner 2022)	Rec	1.20	94.51
	$\text{DH-SNN}_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024)	Rec	0.13	93.80
	$\text{TC-LIF}_{\text{AAAI, 2024}}$ (Zhang et al. 2024)	Rec	0.20	94.84
	SPRiF (ours)	**Rec**	**0.13**	**94.55 ± 0.31**
SHD	Heterogeneous $\text{SNN}_{\text{Nat. Commun., 2021}}$ (Perez-Nieves et al. 2021)	Rec	0.11	82.50
	$\text{ASRNN}_{\text{Nat. Mach. Intell., 2021}}$ (Yin, Corradi, and Bohté 2021)	Rec	0.14	90.40
	$\text{DH-SNN}_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024)	Rec	0.05	91.34
	$\text{BRF}_{\text{ICML, 2024}}$ (Higuchi et al. 2024)	Rec	0.11	91.70
	$\text{TC-LIF}_{\text{AAAI, 2024}}$ (Zhang et al. 2024)	Rec	0.14	88.91
	$\text{MPS-SNN}_{\text{ICLR, 2025}}$ (Ding et al. 2025b)	Rec	0.39	91.19
	$\text{DGN}_{\text{ICLR, 2026}}$ (Bai, Wang, and Yu 2026)	Rec	0.22	87.78
	SPRiF (ours)	**Rec**	**0.05**	**91.52 ± 0.46**

Table 1: Comparison across five temporal benchmarks. "Rec" denotes recurrent synaptic connections; DMP-SNN uses state-space memory shared across a layer. Parameters are reported in millions (M). SPRiF accuracy values are reported as mean±standard deviation over five seeds.

The evidence comprises three components. Table 1 places SPRiF in the context of published benchmark results for gated, adaptive, dendritic, resonate-and-fire, and other temporal neuron dynamics. Controlled ablations and SI-DMS evaluate specific SPRiF mechanisms. Diagnostics based on state space and impulse responses characterize the learned dynamics. The technical appendix specifies preprocessing, architectures, optimization, initialization, and reproducibility details.

Benchmark Comparison

Our five-seed mean exceeds the listed reported accuracies on S-MNIST, PS-MNIST, and QTDB (Table 1). Its absolute margins over the best reported baseline accuracies are 0.08, 0.36, and 1.43 percentage points, respectively. On GSC and SHD, SPRiF is within 0.29 and 0.18 percentage points of the best reported result. The parameter count of the complete model is lower than that of the baseline with the highest reported accuracy on S-MNIST, GSC, and SHD and differs only slightly on PS-MNIST and QTDB. These results provide context across published studies with different protocols. Matched statistical significance and hardware efficiency are outside this comparison.

Contribution of the SPRiF Design Choices

We isolate three design choices on PS-MNIST, GSC, and QTDB. Setting $\omega = 0$ removes rotational coupling while retaining the real decay path. The merged variant combines memory and discharge in a single state with three coordinates, whose first coordinate is both membrane and reset target. Setting $\lambda = 0$ reduces Eq. (11) to scalar reset. All variants otherwise retain the training configuration for each

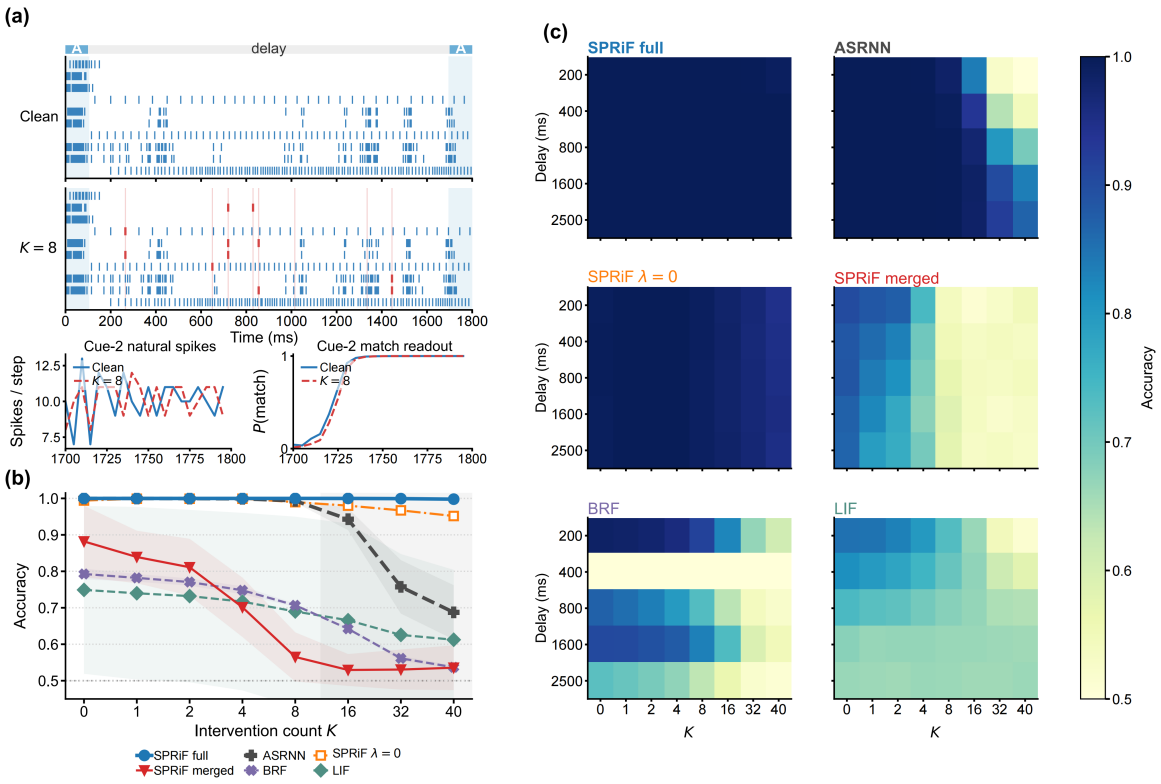


Figure 2: SI-DMS at the fixed 15% intervention fraction. (a) Paired clean and $K = 8$ replays of the same deterministic A–A input for full SPRiF at a 1600-ms delay. The cues share stimulus identity but use independent spike realizations. Blue and red ticks mark natural and forced spikes, and faint red lines mark the eight intervention times. The window for the second cue shows natural activity and the match readout; both conditions predict match. (b) Accuracy versus K, averaged over five delays within each of three seeds; bands show seed standard deviations. (c) Mean accuracy across three seeds for each delay and K, shown with a shared color scale. Training includes $K \leq 8$; $K \in \{16, 32, 40\}$ are unseen. Full SPRiF remains close to its clean accuracy at these unseen intervention counts.

Dataset	Full*	ω=0	Merged	λ=0
PS-MNIST	**95.86**	93.44±0.27	92.29±0.36	95.32±0.18
GSC	**94.55**	93.83±0.28	90.05±0.42	94.22±0.22
QTDB	**88.43**	87.40±0.25	83.39±0.44	87.78±0.30

*Full standard deviations are reported in Table 1.

Table 2: Classification accuracy (%) of SPRiF and its mechanism variants on PS-MNIST, GSC, and QTDB. Full SPRiF is highest and the merged variant is lowest on all three datasets.

task.

Table 2 shows the same ordering on all three datasets. Merging produces the largest observed decrease, 3.57–5.04 percentage points. Removing rotation reduces accuracy by 0.72–2.42 points, with the largest difference on PS-MNIST; scalar reset reduces it by 0.33–0.65 points. The latter two controlled changes therefore have smaller effects under the tested settings. Because the merged variant also has a lower state dimension, its result reflects the combined effects of state decomposition and capacity.

Spike-Intervention Delayed Match-to-Sample (SI-DMS)

Standard benchmark accuracy measures task performance but leaves context corruption induced by spikes unresolved. SI-DMS tests this failure mode by pairing a clean replay with a stressed replay in which forced spikes trigger native reset and recurrent propagation during the memory delay.

The test uses a recurrent hidden layer with 64 units. During the delay, 15% of hidden neurons are induced to spike at each of K sampled timesteps before a match/non-match probe. Placing the membrane coordinate above threshold before spike computation triggers native reset, and the emitted spike enters the recurrent pathway at the next timestep. SI-DMS therefore measures the joint effect of reset triggered by spikes and recurrent spike propagation. Training samples $K \in \{0, 1, 2, 4, 8\}$; evaluation adds unseen $K \in \{16, 32, 40\}$ and sweeps the intervention fraction from 5% to 50%.

For each seed and delay, $\Delta\mathrm{Acc}(K, f) = \mathrm{Acc}(0) -$

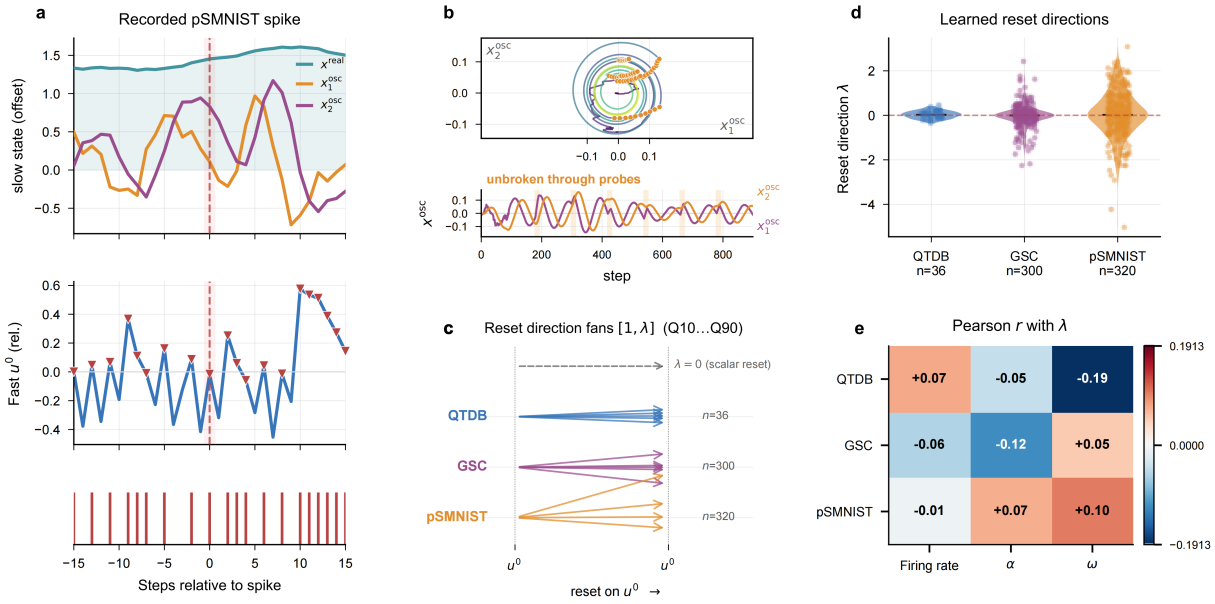


Figure 3: Dynamics of the slow and fast states and learned projective reset. (a) Slow and fast states around a recorded PS-MNIST spike. (b) Oscillatory trajectories of the slow state across cue–delay probe windows. (c) Representative reset directions $[1, \lambda]$. (d) Distributions of λ by task. (e) Pearson correlations between λ and firing rate, α, and ω. Learned reset directions span both signs and are only weakly correlated with the measured neuron statistics.

$\mathrm{Acc}(K, f)$. Table 3 averages over five delays within each seed and then over three seeds at $f = 15\%$. ASRNN provides the control matched in clean accuracy: full SPRiF and ASRNN both attain 100.0 ± 0.0 clean accuracy, but their losses at unseen $K = 40$ are 0.2 ± 0.2 and 31.2 ± 7.4 percentage points, respectively. Clean accuracy is reported alongside paired loss to show the lower starting points of BRF and LIF and the seed variability of LIF.

At $K = 40$, the λ=0 variant loses 4.2 ± 0.2 percentage points, compared with 0.2 ± 0.2 for full SPRiF. This result supports a benefit from the learned reset direction in this task. The merged model has losses of 32.3 ± 1.4 at $K = 8$ and 36.4 ± 2.7 at $K = 40$, alongside lower clean accuracy. Figure 2 shows accuracy across delays and K values; the appendix provides the fraction sweep and implementation checks.

Figure 2(a) replays the same deterministic A–A input under clean and $K = 8$ conditions. After the 1600-ms delay and eight forced events, the stressed replay retains natural activity during the second cue and the same correct match decision. This trace illustrates the intervention process; Figs. 2(b)–(c) provide the aggregate evidence across models, delays, and seeds.

Reset Geometry and Temporal Kernels

The state-space diagnostics connect the formal dynamics to intervention behavior. Figure 3(a) records slow and fast states around a PS-MNIST spike, and Fig. 3(b) follows oscillations of the slow state across cue–delay probe windows. These traces visualize the continuous evolution of the slow state;

Model	Clean	$\Delta\mathrm{Acc}_8$	$\Delta\mathrm{Acc}_{40}$
SPRiF (full)	100.0 ± 0.0	$\mathbf{0.0\pm0.0}$	$\mathbf{0.2\pm0.2}$
SPRiF λ=0	99.4 ± 0.0	0.4 ± 0.1	4.2 ± 0.2
SPRiF merged	88.1 ± 1.4	32.3 ± 1.4	36.4 ± 2.7
ASRNN	100.0 ± 0.0	0.6 ± 0.5	31.2 ± 7.4
BRF	79.3 ± 1.4	8.6 ± 1.4	25.6 ± 1.2
LIF	74.9 ± 23.0	5.9 ± 8.2	13.7 ± 10.2

Table 3: SI-DMS clean accuracy (%) and paired intervention loss (percentage points) at $f = 15\%$, reported as mean±standard deviation over three seeds. For each seed, the value is averaged over five delays before aggregation. Full SPRiF has the smallest loss at the unseen $K = 40$ setting.

Figs. 3(c)–(e) characterize the learned projective reset directions.

The learned reset directions are heterogeneous. On QTDB, GSC, and PS-MNIST, λ spans both signs (Figs. 3(c)–(d)), and its correlations with firing rate, α, and ω remain weak, with $|r| \leq 0.191$ (Fig. 3(e)). Combined with the λ=0 ablation, these distributions indicate that the trained models use heterogeneous reset directions in the fast state.

The impulse response analysis complements SI-DMS by exposing the intrinsic slow transition alongside recall under induced spikes. We inject a unit impulse into trained neurons and record 100 steps of slow state evolution without subsequent input, isolating the temporal basis $A^k B$ in Eq. (10).

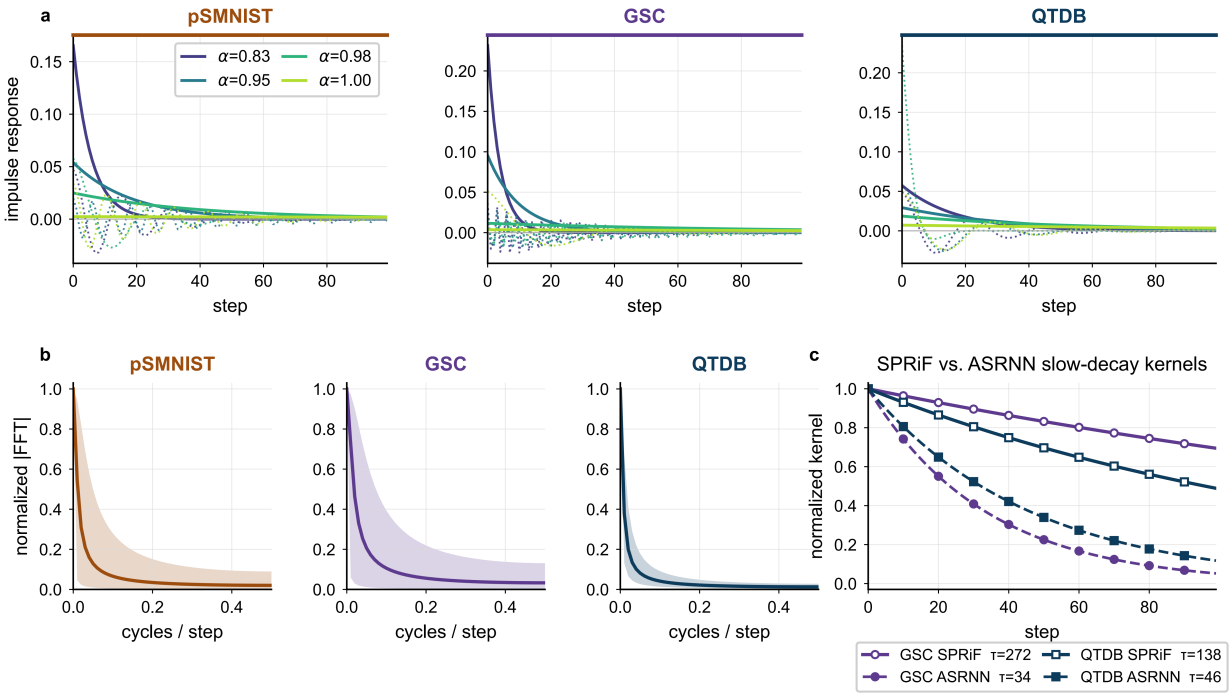


Figure 4: Learned temporal kernels. (a) Real (solid) and oscillatory (dotted) coordinates of four SPRiF neurons selected by α quantile for each task over 100 timesteps after the impulse. (b) Median normalized spectrum of the real coordinate with the 10–90% range. (c) Representative kernels of the SPRiF real coordinate and membrane kernels from the ASRNN checkpoints available for GSC and QTDB, with derived time constants τ. The learned SPRiF basis contains both rapid and persistent responses.

Figure 4 combines trajectories sampled across α quantiles with spectra of the real coordinate aggregated over all neurons.

The real coordinates stratified by α span rapid and persistent responses, and the rotational coordinates vary in phase and frequency. At the population level, the median normalized spectra of the real coordinate concentrate at low frequencies, with substantial variation within each task in the 10–90% ranges. Normalization removes absolute amplitude from this comparison. The resulting response families reflect the constrained parameterization: α controls monotone retention, and ρ and ω govern rotational damping and phase evolution.

We compare kernels across models on GSC and QTDB, the two tasks with available checkpoints. On these tasks, selected SPRiF real modes with slow decay have derived time constants of approximately 272 and 138 steps, respectively, compared with 34 and 46 steps for the trained ASRNN membrane kernels. The appendix provides additional impulse response and spectral analyses.

Conclusion

SPRiF combines a constrained spectral slow state with a resettable fast discharge state and a learned projective reset. Across five temporal benchmarks, SPRiF has the highest reported mean accuracy among the compared methods on three tasks and remains competitive on the other two, with comparable or lower parameter counts than the strongest reported baselines.

At the unseen $K = 40$ setting in recurrent SI-DMS, full SPRiF loses 0.2 ± 0.2 percentage points, compared with 31.2 ± 7.4 for ASRNN matched in clean accuracy and 4.2 ± 0.2 for the $\lambda{=}0$ variant. The equations characterize the intrinsic slow transition and the direct reset path at the current timestep. Controlled ablations support the tested components, while SI-DMS measures their behavior in a synthetic setting with intervention training. Evaluations at larger scale and on neuromorphic hardware remain future work.

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